

TORUN, Poland

WMO: 122500

Lat: 53.0422N

Lon: 18.5953E

Elev: 72

StdP: 100.46

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.7	-12.3	-18.3	0.8	-15.0	-15.3	1.0	-11.7	7.9	4.7	7.1	3.1	1.4	60	0.514

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	30.5	19.7	28.5	18.9	26.7	18.1	21.1	27.8	20.1	26.1	19.2	24.6	3.3	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.0	13.9	23.2	18.1	13.1	22.5	17.2	12.4	21.6	61.2	27.9	58.0	26.1	54.9	24.5	25.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.5	5.8	5.2	DB	-19.0	33.8	5.1	1.5	-22.6	34.9	-25.6	35.8	-28.5	36.7	-32.2	37.8	(4)
(5)			WB	-19.3	22.8	5.0	1.1	-22.9	23.6	-25.8	24.3	-28.6	24.9	-32.3	25.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.8	-1.6	-0.3	3.1	8.6	13.4	16.7	19.1	18.7	14.0	8.8	4.0	0.2		
	DBStd	8.39	5.49	5.20	4.37	4.23	4.18	3.55	3.28	3.15	3.40	3.88	3.85	4.92		
	HDD10.0	1508	359	288	216	74	13	0	0	0	4	69	181	303		
	HDD18.3	3622	617	521	472	292	162	72	29	34	137	294	429	562		
	CDD10.0	1067	0	0	2	33	119	202	283	269	124	33	1	0		
	CDD18.3	140	0	0	0	1	9	24	54	46	6	0	0	0		
	CDH23.3	1461	0	0	0	22	122	275	549	431	62	0	0	0		
	CDH26.7	414	0	0	0	4	21	77	182	118	13	0	0	0		
(14) Wind	WSAvg	2.6	2.9	2.8	2.8	2.7	2.5	2.5	2.4	2.2	2.3	2.5	2.7	2.8		
(15) Precipitation	PrecAvg	538	27	24	27	32	53	72	81	63	49	37	38	37		
	PrecMax	846	57	54	51	103	124	299	237	159	155	130	82	78		
	PrecMin	312	2	3	6	5	14	21	17	3	7	5	11	7		
	PrecStd	109	14	12	12	20	25	49	50	30	32	25	15	18		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.0	11.2	17.2	26.2	29.0	32.5	33.6	32.3	28.5	21.3	13.9	11.1		
		MCWB	8.0	8.1	10.7	15.3	18.0	20.7	20.5	20.6	18.2	16.0	11.9	9.9		
	2%	DB	7.9	9.5	14.3	22.4	26.6	28.7	30.8	29.6	24.9	18.7	12.0	9.2		
		MCWB	6.5	7.3	9.0	13.5	17.0	18.9	19.9	19.7	17.0	14.4	10.3	8.0		
	5%	DB	6.3	7.9	11.8	19.3	24.3	26.4	28.7	27.6	22.5	16.6	10.5	7.7		
		MCWB	5.1	6.1	8.0	11.9	16.0	17.6	19.2	18.9	15.9	13.4	9.1	6.6		
	10%	DB	5.0	6.3	9.5	16.7	21.8	24.3	26.4	25.7	20.3	14.8	9.3	6.3		
		MCWB	3.9	4.7	6.6	10.6	14.6	16.8	18.3	17.9	15.1	12.4	8.0	5.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.5	9.1	11.7	16.0	19.1	21.7	22.3	22.2	19.3	16.7	12.3	10.2		
		MCDB	9.5	10.3	15.8	24.3	26.5	29.3	29.3	29.6	25.8	19.9	13.4	10.8		
	2%	WB	6.6	7.6	9.8	14.2	17.9	20.1	21.2	20.8	18.0	15.0	10.6	8.2		
		MCDB	7.8	9.2	13.2	21.0	24.6	26.5	27.8	27.3	23.3	17.6	11.7	9.0		
	5%	WB	5.3	6.2	8.3	12.6	16.7	18.9	20.3	19.9	16.8	13.7	9.2	6.6		
		MCDB	6.3	7.8	11.1	18.1	22.7	24.6	26.2	25.4	20.7	16.1	10.4	7.5		
	10%	WB	3.9	4.8	6.9	11.2	15.4	17.6	19.3	18.9	15.7	12.6	8.1	5.4		
		MCDB	4.9	6.2	9.3	15.7	20.7	22.6	24.5	23.9	19.4	14.6	9.2	6.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.6	6.0	7.9	10.8	11.3	10.8	10.5	10.9	9.6	7.7	4.8	4.3		
		MCDBR	4.8	7.5	11.7	15.5	14.6	14.5	14.2	14.0	13.1	10.3	6.4	5.1		
	5% WB	MCWBR	4.2	5.7	7.4	8.0	6.6	6.3	5.3	5.9	6.4	6.6	5.1	4.5		
		MCWBR	4.8	7.0	10.3	13.3	12.7	12.3	11.9	12.0	11.0	9.3	6.2	5.0		
(40) Clear-Sky Solar Irradiance	taub		0.311	0.340	0.373	0.420	0.414	0.407	0.441	0.429	0.386	0.365	0.347	0.309		
		taud	2.257	2.264	2.239	2.163	2.236	2.286	2.210	2.246	2.341	2.382	2.325	2.287		
	Ebn at Noon		652	750	802	807	836	845	805	792	784	718	606	583		
		Edh at Noon	64	86	108	133	130	125	133	121	98	77	61	53		
(44) All-Sky Solar Radiation	RadAvg		0.63	1.32	2.56	4.16	5.23	5.45	5.00	4.41	3.16	1.72	0.70	0.45		
	RadStd		0.11	0.21	0.35	0.54	0.53	0.50	0.59	0.40	0.41	0.26	0.09	0.06		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.37	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (29 neighbors)		+0.49	N/A	N/A	N/A	N/A	N/A	N/A	-166	+80	N/A

Nomenclature: See separate page